Iyed MEDIOUNI



SOFTWARE ENGINEER · FULL-STACK, API INTEGRATION & APPLIED AI

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PROFILE

Software engineer. I design and ship complete web platforms - customer journey, online payment, business back-office - and integrate third-party systems through their APIs. I have also carried applied-AI projects into production, including an enterprise assistant that my employer sold to a client. Comfortable from development in React, Node.js and Python through to deployment and the tooling the business teams actually use.

EXPERIENCE

02/2026 - 07/2026
Lac 1, Tunis
Final-year project

AI Engineer - TELNET Holding · AI assistant with fine-tuning

- Project information was scattered across three third-party systems: delivered an enterprise assistant on a LangGraph agent and a Neo4j knowledge graph that queries Jira, Confluence and GitLab through their APIs and answers in natural language from one interface. TELNET sold the product to a client.
- Cut response time 2.4x (7.3s to 3.0s) through deterministic query routing and API calls executed in parallel.
- Built the quality gate: every answer is scored on two axes, and a fine-tuned model ships only after a statistical test on held-out data - the adapter was promoted at 2.1 standard errors on 366 questions, with 94.8% of answers grounded in a cited source.
- Shipped ready to operate: Docker-containerised FastAPI services, MongoDB persistence, local inference via Ollama, MLflow tracking each training run, and a React/Vite console for day-to-day control, with quality, latency and failures monitored in production.

PROJECTS

09/2025 - 11/2025

Multi-Agent Automation & API Integration · github.com/iyed122/automated-twitter-content

- Generic output drifts from what is actually trending: pipeline of autonomous agents (CrewAI + LLaMA-3-70B, Groq) that gathers data from several online sources, cross-checks it, then produces content tied back to those sources.
- Full third-party API integration with secure OAuth authentication, from collection through to automated publication.

07/2025 - 10/2025

Autonomous Vehicle Perception · github.com/iyed122/autonomous-vehicule-VLM

- End-to-end pipeline on AWS SageMaker pairing a fine-tuned YOLOv8 detector with a custom bird's-eye-view model for trajectory prediction (PyTorch, ADE 0.8 / FDE 1.2).
- Streamlit dashboard surfacing live detection and trajectories, with a vision-language model (Qwen2.5-VL) and text-to-speech speaking the verdict aloud - keep going, take a break, or a vehicle is close.

02/2023 - 06/2023
Ghazela, Technopole
Final-year project

Full-Stack & AI Developer - AURAX · E-commerce platform development with AI

- A catalogue alone does not convert: delivered the entire customer journey in MERN (React, Node.js, Express, MongoDB) - search and filters, cart, secure online payment via Stripe, promotion codes, confirmation, order tracking and history.
- Making every change depend on a developer is expensive: built a self-service back-office where the business team manages catalogue, pricing, promotions and stock, with a revenue and orders dashboard.
- White-label customisation: logo, colours and the site announcement banner editable by the administrator with no technical involvement.
- Content-based recommendation engine (VGG16 to PCA to KNN) trained over 50,000+ product images and exposed through a Flask API, working from a user's first visit with no history.
- Delivered inside the company under production practices: JWT token authentication, code versioned on GitLab, tasks tracked in Linear.

EDUCATION

2023 - 2026
Engineering Degree - Data Science & Artificial Intelligence
TEK-UP University

2020 - 2023
Bachelor in Computer Science
ISAMM, Manouba

TECHNICAL SKILLS

WEB DEVELOPMENT	React · Vite · Node.js · Express · Flask · FastAPI · JavaScript · REST
INTEGRATION & API	Third-party APIs (Jira, Confluence, GitLab, X) · OAuth · Stripe · JWT · Postman · parallel calls & routing
DATA	MongoDB · Neo4j · PostgreSQL · MySQL · Qdrant · Cassandra · SQL / PL/SQL · Power BI
AI & MACHINE LEARNING	LLMs · LangGraph · LangChain · CrewAI · RAG & Graph-RAG · embeddings · RAGAS · RAFT fine-tuning · Ollama · PyTorch · TensorFlow · recommender systems
CLOUD & TOOLING	AWS (EC2, S3, Lambda, SageMaker) · Docker · MLflow · Git / GitLab · Linear

LANGUAGES & CERTIFICATIONS

ARABIC	Native
FRENCH	Professional working proficiency - academic and technical writing
ENGLISH	IELTS Academic, British Council - B2 (6.5 / 9)